Green V2X: An Energy-Aware Handoff Strategy for Reducing Uplink Transmission Energy and CO_2 Emissions in Vehicular Networks

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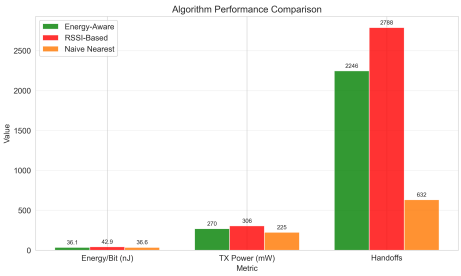
Abstract:

Vehicle-to-Everything (V2X) communication is central to next-generation intelligent transportation systems, where moving vehicles continuously maintain uplink connections with roadside cellular base stations (BSs). As a vehicle traverses a highway, it must periodically switch its serving BS — a process known as *handoff*. Conventional handoff policies select the BS with the strongest received signal strength indicator (RSSI) or highest signal-to-interference-plus-noise ratio (SINR), with no regard for the energy cost of the chosen radio link. This energy-blind approach leads to avoidable transmit power expenditure and unnecessary CO_2 emissions — a growing concern as vehicular networks scale toward millions of connected vehicles worldwide.

This work presents **Green V2X**, a research simulation framework that introduces and evaluates an **Energy-Aware Handoff (EAH)** algorithm for highway V2X uplink scenarios. The EAH algorithm scores each candidate BS using a composite metric that minimises *Energy Per Bit* (EPB) — joules consumed per bit successfully delivered — while jointly accounting for BS load (preferring under-utilised cells), Quality-of-Service (QoS) constraints (minimum data rate and SINR thresholds), and a cell-edge stability penalty. Switching decisions employ hysteresis with a time-to-trigger (TTT) timer and per-vehicle cooldown periods to suppress ping-pong oscillations; forced-trigger rules override hysteresis when the current BS is severely overloaded or the link degrades beyond acceptable bounds. The simulator models realistic highway mobility (multi-lane, lane-dependent speeds, stochastic lane changes), a full physical-layer channel (path loss, weather attenuation, shadowing, fast fading, inter-cell interference), a PA-level energy model, and a CO_2 conversion pipeline calibrated to the Bangladesh electricity grid carbon intensity. Seven classical and literature-inspired baseline algorithms are benchmarked under identical, seed-controlled conditions for a rigorous and fair comparison.



Algorithm	Avg. EPB (J/bit)	Outage (%)
EAH (Proposed)	$\mathbf{3.61 \times 10^{-8}}$	**16.37**
RSSI	4.30×10^{-8}	22.75
SINR	4.14×10^{-8}	∼21.0
Load-Aware RSSI	4.31×10^{-8}	∼23.0
Naive Nearest	3.66×10^{-8}	∼18.0
Literature UL-HO	4.31×10^{-8}	24.75
LB-Aware RSRP	4.34×10^{-8}	∼23.0
MDPI Energy Efficient	5.18×10^{-8}	68.25

15-seed average; **bold** = best.

Figure 1: (**Left**) Mean Energy Per Bit (EPB) comparison across all eight handoff algorithms (15-seed average, highway scenario, clear weather). (**Right**) Mean EPB and outage by algorithm; EAH achieves the lowest EPB while maintaining service quality superior to all baselines.

Across 15 independent simulation seeds (20 vehicles, 16 BSs, 1,200 s, clear weather), EAH achieves a mean EPB saving of **15.26 %** $\pm$ **1.07 %** over RSSI ($p \approx 10^{-18}$, paired t-test) and a total energy saving of **9.09 %** $\pm$ **0.55 %**, directly reducing CO_2 by the same margin. As shown in Fig. 1, EAH records the lowest EPB among all algorithms while simultaneously achieving the best outage probability (16.37 %) and highest mean throughput (24.27 Mbps). Handoff count falls by **20.52 %** and ping-pong rate drops from 34.54 % to 20.46 %. The MDPI-inspired baseline achieves very few handoffs but suffers severe service collapse (outage 68.25 %, throughput 8.29 Mbps), confirming that minimising switching alone does not produce a quality link. Scalability tests (20–200 vehicles) show ≈**23 %** stable savings, and weather sweeps reveal amplified gains under adverse conditions — heavy rain **49 %**, thunderstorm **69 %** — precisely when energy-blind policies waste the most power.

These results demonstrate that handoff policy is an effective and largely untapped lever for green vehicular networking. By explicitly minimising EPB rather than maximising signal strength, EAH simultaneously reduces radio energy consumption, lowers CO_2 emissions, and improves service reliability — without any hardware modification. The findings are directly relevant to sustainable transportation and climate-resilient smart-city infrastructure, contributing to global efforts to reduce the carbon footprint of wireless communication systems.

Keywords: V2X Communication; Energy-Aware Handoff; Green Wireless Networks; CO_2 Reduction; Sustainable Transportation